

Geometric Sequences and Series

Multiply by the same r every time. Divide terms to find r , use logs when n is in the power, and only add to infinity when $|r| < 1$.



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THE TOPIC ON ONE MAP

Geometric Sequences and Series

01

The n th term

$$ar^{n-1}$$

divide two terms

middle² = product

02

The sum S_n

$$\frac{a(1-r^n)}{1-r}$$

smallest n : logs

prove the formula

03

Sum to infinity

$$\frac{a}{1-r}$$

$$|r| < 1$$

choose the right r

THE THREE FORMULAS

$$u_n = ar^{n-1} \quad S_n = \frac{a(1-r^n)}{1-r} \quad S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

n in the power $\Rightarrow$ take logs $\cdot$ log of a number less than 1 is negative: the inequality flips

01 The n th Term: $u_n = ar^{n-1}$

Divide one term by another to cancel a and leave a power of r . Three consecutive terms in algebra: (middle)² = first $\times$ third.

WHEN YOU SEE $\rightarrow$ YOU DO

“the 3rd term is 20 and the 5th term is 12.8”

$$\rightarrow \frac{ar^4}{ar^2} = r^2$$

“ $2k-15$, k , $k+4$ are consecutive terms”

$$\rightarrow k^2 = (2k-15)(k+4)$$

“the n th term is greater than 10^{100} ”

$$\rightarrow 4 \times 6^{n-1} > 10^{100}: \text{logs}$$

“is 8192 a term?”

$$\rightarrow \text{solve } ar^{n-1} = 8192: \text{ is } n \text{ a whole number?}$$

WORKED MODEL • JUNE 2023

Downloads $2k-15$, k , $k+4$ (thousands) are geometric. Show $k^2 - 7k - 60 = 0$.

$$1 \quad \frac{k}{2k-15} = \frac{k+4}{k}$$

same ratio

$$2 \quad k^2 = 2k^2 - 7k - 60$$

cross-multiply

ANSWER $k^2 - 7k - 60 = 0 \Rightarrow k = 12 \ (k > 7.5)$

IDEA 01

Practise: The n th Term: $u_n = ar^{n-1}$

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Divide the terms

P2 · TOPIC 6 3 marks

The 3rd term of a geometric sequence is 12 and the 6th term is 96. Find a and r .

$$\frac{ar^5}{ar^2} = r^3.$$

Q2 DR ESLAM Algebraic terms

P2 · TOPIC 6 3 marks

x , $x + 4$, $4x$ are consecutive terms of a geometric sequence, $x > 0$. Find x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 6**4 marks**

1 Identify which of the following are geometric sequences. For those that are, write down the first term and the common ratio.

(i) 3, 8, 13, 18, ...

(ii) 5, 15, 45, 135, ...

(iii) 5, -10, 20, -40, ...

(iv) $\frac{1}{3}, \frac{1}{6}, \frac{1}{12}, \dots$

(4 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 6 6 marks

- 1 (a) The first three terms of a geometric sequence are given by x^2 , $4x$, and $x + 14$ respectively, where $x > 0$.

Show that $x^3 - 2x^2 = 0$.

(2 marks)

- (b) Find the value of the 15th term of the sequence.

(3 marks)

- (c) State, with a reason, whether 8192 is a term in the sequence.

(1 mark)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 6**5 marks**

- 1 The first three terms of a geometric sequence are given by $x + 11$, $5x$, and $3x^2$ respectively, where x is a non-zero real number.

Find the value of the sixth term in the sequence, giving your answer as a fraction.

(5 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · June 2023 · Q6

P2 · TOPIC 6 9 marks

6. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$ (2)

(b) predict the number of downloads in the 4th month. (4)

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the N th month.

(c) Calculate the value of N according to the model. (3)

(Total for Question 6 is 9 marks)

02 The Sum S_n and the Smallest n

Write S_n with the numbers in, make r^n the subject, then take logs of both sides. Round n **up** for “greater than”. Dividing by $\log r < 0$ flips the inequality.

WHEN YOU SEE → YOU DO

“the sum of the first n terms is greater than 156”

→ $S_n > 156$, isolate r^n , logs

“smallest possible value of n ”

→ solve, then round up; check n and $n - 1$

“prove that $S_n = \frac{a(1-r^n)}{1-r}$ ”

→ write S_n and rS_n , subtract

“8% less each year”, “150 litres at the start”

→ $r = 0.92$, $a = 150$

“solutions based entirely on graphical or numerical methods are not acceptable”

→ show the log step

Write S_n

Isolate r^n

Take logs

Round up

WORKED MODEL • JANUARY 2023

$a = 31.25$, $r = 0.8$. Smallest n with $S_n > 156$.

$$1 \quad \frac{31.25(1-0.8^n)}{0.2} > 156 \Rightarrow 0.8^n < 0.0016$$

isolate r^n

$$2 \quad n > \frac{\log 0.0016}{\log 0.8} = 28.8$$

logs — the sign flips

ANSWER $n = 29$

IDEA 02

Practise: The Sum S_n and the Smallest n

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM** A sum

P2 · TOPIC 6 2 marks

Find the sum of the first 10 terms of $2 + 6 + 18 + \dots$

Q8**DR ESLAM** Smallest n with logs

P2 · TOPIC 6 4 marks

Find the smallest n for which $3 + 6 + 12 + \dots$ (to n terms) exceeds 1000.

SELECTED PRACTICE

Q9**PRACTICE** Practice · Easy**P2 · TOPIC 6** 4 marks**2 (a)** Evaluate

$$\sum_{r=1}^5 3(2^r)$$

(2 marks)**(b)** Evaluate.

$$\sum_{r=4}^8 (-1)^r(2^r)$$

(2 marks)

Q10**PRACTICE**

Practice · Hard

P2 · TOPIC 6 5 marks

- 2 A geometric series has first term 14 and common ratio $\frac{99}{100}$.

Given that the sum of the first k terms of the series is less than 1000, find the largest possible value of k .

(5 marks)

Q11**PRACTICE**

Practice · Very Hard

P2 · TOPIC 6**4 marks**

- 4 A geometric series has first term a , and its terms are connected by the relationship $u_{n+4} = 9u_n$ for all $n \geq 1$.

Given that all the terms of the series are positive, show that the sum of the first twelve terms of the series may be written in the form

$$S_{12} = ka(\sqrt{n} + 1)$$

where k and n are positive integers and $\sqrt{n}$ is a surd.

(4 marks)

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA12 · January 2023 · Q7

P2 · TOPIC 6 7 marks

7. A geometric sequence has first term a and common ratio r , where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

(a) show that $r = 0.8$

(1)

(b) Hence find the value of a .

(2)

Given that the sum of the first n terms of this sequence is greater than 156

(c) find the smallest possible value of n .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

(Total for Question 7 is 7 marks)

Q13**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q8

P2 · TOPIC 6 8 marks

8. (i) A geometric series has first term a and common ratio r .

Prove that the sum of the first n terms of this series S_n is given by

$$S_n = \frac{a(1-r^n)}{1-r} \quad (3)$$

- (ii) A liquid is to be stored in a barrel.

Due to evaporation, the volume of the liquid in the barrel at the end of each year is 8% less than the volume of the liquid in the barrel at the start of the year.

At the start of the first year, the barrel is filled with 150 litres of the liquid.

- (a) Show that the amount of the liquid in the barrel at the end of 6 years is approximately 91 litres. (2)

At the start of each year a new barrel is filled with 150 litres of the liquid so that, at the end of 40 years, there are 40 barrels containing the liquid.

- (b) Calculate the total amount of the liquid, to the nearest litre, in the 40 barrels at the end of 40 years. (3)

(Total for Question 8 is 8 marks)

03 Sum to Infinity: $|r| < 1$

A sum to infinity exists only when $-1 < r < 1$. When a quadratic gives two values of r , the condition chooses one. $S_{\infty} = \frac{a}{1-r}$ together with a term or a finite sum gives two equations.

WHEN YOU SEE → YOU DO

“given that the series has a sum to infinity”

→ reject $|r| \geq 1$

“ $S_{\infty} = 3a$ ”

→ $\frac{a}{1-r} = 3a \Rightarrow r = \frac{2}{3}$

“the second term is -6 , $S_{\infty} = 25$ ”

→ $ar = -6$, $\frac{a}{1-r} = 25$: eliminate a

“ $S_3 = 70.2$ and $S_{\infty} = 75$ ”

→ $\frac{S_3}{S_{\infty}} = 1 - r^3$

“is convergent”, $-1 + 3x - 9x^2 + \dots$

→ $|r| = |-3x| < 1$

WORKED MODEL · JANUARY 2025

$S_3 = 70.2$ and $S_{\infty} = 75$. Find r and a .

$$1 \quad \frac{a(1-r^3)}{1-r} \div \frac{a}{1-r} = 1 - r^3 = \frac{70.2}{75} = 0.936$$

divide

$$2 \quad r^3 = 0.064 \Rightarrow r = 0.4$$

ANSWER $a = 75(1 - 0.4) = 45$

× **TRAP** Keeping $r = \frac{6}{5}$ when the series has a sum to infinity. $|r|$ must be less than 1.

IDEA 03

Practise: Sum to Infinity: $|r| < 1$

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Sum to infinity

P2 · TOPIC 6 3 marks

(a) Find S_{∞} for $24 + 12 + 6 + \dots$ (b) A series has $a = 5$ and $S_{\infty} = 20$. Find r .

SELECTED PRACTICE**Q15** PRACTICE Practice · Easy

P2 · TOPIC 6 3 marks

8 (a) The first term of a geometric sequence is 6.

The sum to infinity is 8.

Show that the common ratio is 0.25.

(2 marks)

(b) Briefly explain why the geometric sequence with first term 6 and common ratio 0.25 has a sum to infinity.

(1 mark)

Q16**PRACTICE**

Practice · Hard

P2 · TOPIC 6 8 marks

- 6 (a)** The second and fifth terms of a geometric series are 13.44 and 5.67 respectively. The series has first term a and common ratio r .

By first determining the values of a and r , calculate the sum to infinity of the series.

(6 marks)

- (b)** Calculate the difference between the sum to infinity of the series and the sum of the first 20 terms of the series. Give your answer accurate to 2 decimal places.

(2 marks)

Q17**PRACTICE**

Practice · Very Hard

P2 · TOPIC 6**13 marks**

- 7 (a)** The geometric series $S = u_1 + u_2 + u_3 + \dots + u_n + \dots$ is convergent, and the sum to infinity of the series is S_∞ . The first term of the series is a , and the common ratio is r .

A different series $T = u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2 + \dots$ is formed by squaring all the terms of the series S above.

Show that $T = u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2 + \dots$ is also a convergent geometric series.

(4 marks)

- (b)** The sum to infinity of the series T is T_∞ .

Express the ratio $\frac{T_\infty}{S_\infty}$ in terms of a and r , simplifying your answer as far as possible.

(3 marks)

- (c)** Show that if $T_\infty = S_\infty$, then $u_k^2 = u_{2k-1} + u_{2k}$ for all $k \geq 1$. Comment on what this shows about the relationship between the terms of the two series.

(6 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA12 · October 2021 · Q7

P2 · TOPIC 6 10 marks

7. (i) A geometric sequence has first term 4 and common ratio 6
Given that the n^{th} term is greater than 10^{100} , find the minimum possible value of n . (3)

- (ii) A different geometric sequence has first term a and common ratio r .

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

- (a) show that

$$25r^2 - 25r - 6 = 0 \quad (3)$$

- (b) Write down the solutions of

$$25r^2 - 25r - 6 = 0 \quad (1)$$

Hence,

- (c) state the value of r , giving a reason for your answer, (1)

- (d) find the sum of the first 4 terms of the series. (2)

(Total 10 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q6

P2 · TOPIC 6 8 marks

6. In a geometric sequence $u_1, u_2, u_3, \dots$

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

(a) Show that r satisfies

$$3r^2 - 4r - 4 = 0$$

(3)

Given that the geometric sequence has a sum to infinity,

(b) find u_1

(3)

(c) find S_∞

(2)

(Total 8 marks)

Q20 **PAST PAPER** Pearson IAL WMA12 · October 2022 · Q8**P2 · TOPIC 6** **7 marks**

8. A geometric sequence has first term a and common ratio r

Given that $S_{\infty} = 3a$

- (a) show that $r = \frac{2}{3}$ (2)

Given also that

$$u_2 - u_4 = 16$$

where u_k is the k^{th} term of this sequence,

- (b) find the value of S_{10} giving your answer to one decimal place. (5)

(Total for Question 8 is 7 marks)

Q21**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q4

P2 · TOPIC 6**6 marks**

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

(a) the common ratio, (4)

(b) the first term. (2)

(Total for Question 4 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------------|--|
| 01 | nth term | Divide terms for r ; middle ² = first $\times$ third. |
| 02 | Sum S_n | Isolate r^n , logs, round up (sign flips for $r < 1$). |
| 03 | S_∞ | Only if $ r < 1$; it picks the right root. |

BOTTOM LINE Same ratio $\cdot n$ in the power $\rightarrow$ logs $\cdot$ infinity only when $|r| < 1$.

Answers

Geometric Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE n TH TERM: $u_n = ar^{n-1}$

- Q1** Dr Eslam · Divide the terms $r^3 = 8 \Rightarrow r = 2, a = 3$
- Q2** Dr Eslam · Algebraic terms $(x + 4)^2 = 4x^2 \Rightarrow x + 4 = 2x: x = 4$ (terms 4, 8, 16)
- Q3** Practice · Easy (i) not geometric (ii) $a = 5, r = 3$ (iii) $a = 5, r = -2$ (iv) $a = \frac{1}{3}, r = \frac{1}{2}$
- Q4** Practice · Medium (a) $16x^2 = x^2(x + 14) \Rightarrow x^3 - 2x^2 = 0$ (b) $x = 2: 4, 8, 16; u_{15} = 4(2^{14}) = 65536$ (c) yes: $8192 = 4(2^{11})$, the 12th term
- Q5** Practice · Very Hard $25x^2 = 3x^2(x + 11) \Rightarrow x = -\frac{8}{3}; a = \frac{25}{3}, r = -\frac{8}{5}; u_6 = -\frac{32768}{375}$
- Q6** Pearson IAL WMA12 · June 2023 · Q6 (a) Proven. (b) $\frac{64}{3}$ thousand downloads, about 21300 downloads. (c) $N = 17$

IDEA 02 · THE SUM S_n AND THE SMALLEST n

- Q7** Dr Eslam · A sum $\frac{2(3^{10}-1)}{2} = 59048$
- Q8** Dr Eslam · Smallest n with logs $3(2^n - 1) > 1000 \Rightarrow 2^n > 334.3 \Rightarrow n > 8.38:$
 $n = 9$
- Q9** Practice · Easy (a) $3(2 + 4 + 8 + 16 + 32) = 186$ (b) $16 - 32 + 64 - 128 + 256 = 176$
- Q10** Practice · Hard $1400(1 - 0.99^k) < 1000 \Rightarrow 0.99^k > \frac{2}{7} \Rightarrow k < 124.6: k = 124$
- Q11** Practice · Very Hard $r^4 = 9 \Rightarrow r = \sqrt{3}; S_{12} = \frac{a(729-1)}{\sqrt{3}-1} = 364a(\sqrt{3} + 1)$
- Q12** Pearson IAL WMA12 · January 2023 · Q7 (a) $r = 0.8.$ (b) $a = 31.25.$ (c) $n = 29$
- Q13** Pearson IAL WMA12 · October 2025 · Q8 (i) Proven. (ii)(a) Approximately 91 litres.
(ii)(b) 1664 litres

IDEA 03 · SUM TO INFINITY: $|r| < 1$

- Q14** Dr Eslam · Sum to infinity (a) $\frac{24}{1-\frac{1}{2}} = 48$ (b) $\frac{5}{1-r} = 20 \Rightarrow r = \frac{3}{4}$
- Q15** Practice · Easy (a) $\frac{6}{1-r} = 8 \Rightarrow r = \frac{1}{4}$ (b) $|r| < 1$
- Q16** Practice · Hard (a) $r^3 = 0.421875 \Rightarrow r = 0.75, a = 17.92:$
 $S_{\infty} = 71.68$ (b) $71.68 \times 0.75^{20} \approx 0.23$
- Q17** Practice · Very Hard (a) T has first term a^2 , ratio r^2 , and $r^2 < 1$ (b) $\frac{a}{1+r}$ (c) $a = 1+r$, so $u_{2k-1} + u_{2k} = ar^{2k-2}(1+r) = a^2 r^{2k-2} = u_k^2$
- Q18** Pearson IAL WMA12 · October 2021 · Q7 (i) $n = 129.$ (ii)(a) Proven. (ii)(b) $r = \frac{6}{5}$ or $-\frac{1}{5}.$ (ii)(c) $r = -\frac{1}{5}.$ (ii)(d) $\frac{624}{25}$
- Q19** Pearson IAL WMA12 · June 2022 · Q6 (a) Proven: $3r^2 - 4r - 4 = 0.$ (b) $u_1 = -27.$ (c) $S_{\infty} = -\frac{81}{5}$
- Q20** Pearson IAL WMA12 · October 2022 · Q8 (a) $r = \frac{2}{3}$ (b) $a = 43.2; S_{10} = \frac{92840}{729} \approx 127.4$
- Q21** Pearson IAL WMA12 · January 2025 · Q4 (a) $r = 0.4.$ (b) 45